

## Continuous regeneration of the draw solution in textile wastewater treatment using a combination of simultaneous forward osmosis and reverse osmosis

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### ABSTRACT

The concerning and abundant textile wastewater can be treated by forward osmosis (FO) in order to reduce its volume and simultaneously recover clean water. However, the productivity of FO depends on the concentration of the draw solution that is used. In this work, a simultaneous application of FO and reverse osmosis (RO) is proposed. The HFFO14® FO membrane (Aquaporin, Denmark) was employed to concentrate a real textile wastewater, whereas the SW30-2540 (DuPont, USA) RO membrane was employed to simultaneously regenerate the draw solution, which consisted in a 0.7 M NaCl solution, and to obtain a clean water stream. The concentration of the textile wastewater increased until 90% water recovery was achieved. The rejection values obtained for the chemical oxygen demand and total organic carbon were in the range 99 – 100%. Afterwards, the previously concentrated textile wastewater was again processed until a volume concentration factor of 16.5 was reached. Stable values of permeate flux (around 4 L/h·m<sup>2</sup>) were obtained in the FO process, whereas the reverse osmosis step permitted the maintenance of a stable conductivity in the draw solution and provided clean water as permeate.

### 1. Introduction

The textile sector is one of the major wastewater producers in the industrial context [1]. The increasing manufacturing of clothing and apparel leads to a serious generation of contaminant effluents derived from the dyeing of fabrics and the finishing processes of textile items. Textile wastewater is characterized by its complex and highly variable composition, resulting from a sequence of wet processing operations such as desizing, scouring, bleaching, dyeing, and finishing [2]. Each of these stages derives in a diverse range of pollutants, leading to effluents with elevated pH (surpassing a pH 9 in many cases), high temperature, and substantial loads of total dissolved solids (TDS). The resulting variety of wastewater typically contains high chemical oxygen demand (COD) and is characterized by a significant presence of

non-biodegradable organic compounds. The organic fraction of textile wastewater primarily includes residual dyes, surfactants, sizing agents, and dispersing chemicals [3]. Synthetic dyes, such as azo, reactive, and vat dyes are of particular concern due to their resistance to degradation and potential ecotoxicity. These compounds often contain stable aromatic structures, making them persistent in aquatic environments [4]. Additionally, surfactants and other auxiliaries may contribute to foam formation, COD, and toxicity to microbial communities during treatment. Inorganic constituents are also significant, including salts, like sodium chloride and sodium sulfate used in dye baths. Heavy metals such as copper, chromium, and zinc are present as well [5].

Beyond their chemical complexity, textile effluents imply serious environmental risks. The intense coloration reduces light penetration in water bodies, affecting photosynthesis and aquatic life. The current

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environmental awareness, as well as the water shortages that are becoming more and more common, demand an efficient disposal of these effluents [6]. However, the high variability in the composition of these wastewater further complicates treatment strategies. Conventional biological treatment methods often fall short in removing color and persistent compounds. In many cases, textile dyes are removed from the wastewater by means of physical treatments, (such as coagulation, flocculation, and adsorption), or chemical treatments, including the Fenton reaction or ozonation [7–9]. Even though many of these methodologies are simple and easy to implement, they do not allow the reuse of the chemicals and, in some cases, they may result in toxic intermediates [10]. Nowadays, the principles of the circular economy encourage the recovery of chemicals and water from textile wastewater. The ideal situation would imply the reutilization of both the recovered chemicals and clean water in the following rounds of textile dyeing or finishing, reducing the environmental impact of this industry.

In this context, several membrane processes have been proposed to assess the treatment of these textile effluents. These processes include conventional membrane technologies, such as microfiltration (MF) [11], ultrafiltration (UF) [12–14], nanofiltration (NF) [15–17] and reverse osmosis [18]. Additionally, emergent membrane technologies, such as forward osmosis (FO) have been proposed in recent years. Several authors have previously demonstrated the efficiency of FO in dealing with textile wastewater [19,20]. In fact, the relatively low degree of fouling occurring in FO facilitates the reuse of the membranes, as they are less affected by harsh chemical or mechanical cleaning processes and thus exhibit an extended operational lifespan. This durability can reduce the frequency of replacement and the associated demand for raw materials, aligning with the principles of circular economy.

On the contrary, the main challenge that accompanies FO is the progressive loss of the driving force of the process, as the draw solution becomes more diluted and the osmotic pressure difference between both sides of the membrane decreases. This implies the necessity of regenerating the draw solution, in order to avoid the decrease in the permeate flux. The regeneration of the draw solution represents an additional cost for the process [21] and it is not direct. However, it is of utmost importance for the technical and economic viability of FO.

Most of the studies dealing with FO indicate the use of saline solutions as draw solutions. In those cases, reverse osmosis (RO) can be employed to restore the concentration of the salt, as RO membranes efficiently reject ions. However, the majority of the studies combining FO and RO are not focused on the regeneration of the draw solution. On the contrary, when FO and RO have been combined in the literature, the focus is normally on the desalination of seawater. Therefore, the brine is employed as the draw solution of a FO process and, once this brine is partially diluted, it is later treated by RO [22,23]. The real-time regeneration of the draw solution during the FO process by coupling it with a simultaneous (not sequential) RO process has been far less explored. D'Haese et al. proposed a simultaneous FO-RO system during the treatment of aqueous solutions spiked with organic micropollutants [24] and trace organic compounds [25]. Later, Liu et al. implemented a similar strategy to process synthetic solutions containing pharmaceuticals and personal care products [26]. However, the application of this valuable closed-loop technology to treat challenging, real wastewater (such as textile wastewater) is absent in the literature.

Therefore, the innovation covered by this paper includes the following aspects: treatment of large volumes, industrial perspective, application of a sophisticated closed-loop FO-RO system with real textile wastewater, and the use of a large-area FO membrane which is not commonly used with real effluents. Large volumes of real textile effluents were treated in a pilot plant, trying to obtain results that can be extrapolated to an industrial context. The management of high volumes in a laboratory entails a challenge by itself, because the equipment and materials are not always designed to handle large volumes. Nevertheless, these studies are essential to explore a real perspective on the industrial application of the developed processes. Therefore, real textile

wastewater was processed employing the hollow-fiber Aquaporin® HFFO14 membrane (Kongens Lyngby, Denmark), while the draw solution was continuously regenerated by a coupled RO process. This system allowed the concentration of dyes and salts from the textile wastewater and, at the same time, the obtaining of clean water to be reused in a subsequent stage of textile production.

## 2. Materials and methods

### 2.1. Textile wastewater

The textile effluent studied in this work was obtained from a Danish textile company that manufactures cotton and wool textiles, employing reactive dyes. This wastewater was obtained at the outlet of the equalization tank in this industry; therefore, it was called end-of-pipe wastewater. The equalization tank collects all the wastewater generated in the different stages of textile manufacturing (including dyeing, neutralization, soaping, softening, washing, etc.). Even though the exact composition of this textile effluent was not provided by the company, reactive dyes, salts, emulsifiers, enzymes and surfactants are expected to be present.

The parameters obtained after the characterization of this wastewater are presented in Table 1. As presented in the Table, considerable levels of COD and TDS, as well as high conductivity were observed. This characterization indicated that, as expected, this textile effluent could not be directly discharged into the sewage system and needed to be further treated.

The equalization tank contains all the waste streams generated in the textile industry. This can also include aqueous streams from rinsing steps, pre-washing and post-washing steps, as well as all types of wastewaters from the industry. Therefore, a homogenization of the temperature can be expected. Also, a dilution effect may occur, in comparison with textile wastewater collected right after a dyeing or finishing process. Considering that the objective of the simultaneous FO-RO process is to be installed in a real industry, it seems reasonable to locate the plant right after the equalization tank. In a real scenario, the equalization tank allows the accumulation of wastewater (which can be produced at different flow rates, depending on the type of product manufactured) in order to supply the FO-RO plant with a constant flow rate, avoiding any oscillation [27]. Therefore, in this work, the wastewater coming from the equalization tank was employed to study the membrane performance. The values presented in Table 1 are consistent with those reported for textile effluents. According to previous studies, such wastewaters are typically complex mixtures that contain residual dyes, such as reactive dyes, which contribute to persistent color and recalcitrant organic matter [3]. They also include high concentrations of salts, especially NaCl and Na<sub>2</sub>SO<sub>4</sub>, which are extensively used in dyeing processes to improve dye fixation [6,28]. The resulting high salinity is reflected in the measured conductivity and total dissolved solids. Moreover, textile effluents often present variable organic loads, ranging from low COD values when dominated by dyeing and rinsing operations to very high COD (>1000 mg/L) when finishing agents, surfactants, or auxiliary chemicals are present. The neutral pH observed in our case (7.3) is also in line with equalization tank effluents, where acidic and

**Table 1**

Characterization of the textile wastewater employed in this work. The relative standard deviation of the measurements was always below 18%.

| Parameter                     | Value |
|-------------------------------|-------|
| COD (mg/L)                    | 140   |
| Total dissolved solids (mg/L) | 2583  |
| Conductivity (μS/cm)          | 3394  |
| Osmolality (osmol/kg)         | 0.052 |
| pH                            | 7.3   |

alkaline streams from different process steps are mixed. Although osmolality has rarely been reported in previous works, the values measured in this study are coherent with the observed levels of TDS and chemical composition of such discharges. Altogether, this characterization falls within the expected range for textile wastewater and sets the sample as a proper representative of this type of effluent.

When necessary, the osmotic pressure ( $\pi$ ) of the solutions was estimated according to the Van't Hoff equation:

$$\pi = i \cdot M \cdot R \cdot T \quad (1)$$

being  $i$  the Van't Hoff factor, which represents the number of ions in which a molecule is dissociated;  $M$  represents the molarity of the solution,  $R$  represents the ideal gas constant, and  $T$  represents the temperature of the system.

## 2.2. Pilot plant setup

A pilot plant, which combines an FO line and an RO line, was employed to perform the treatment of the textile wastewater. The recovery of water ( $Recovery_t$ ) during the FO process was calculated ac-

cording to the following equation:

$$Recovery_t = \frac{Feed\ mass_0 - Feed\ mass_t}{Feed\ mass_0} \cdot 100 \quad (2)$$

The employed pilot plant was designed for the simultaneous application of FO and RO. Therefore, this system allows the utilization of an RO retentate (which normally has high osmotic pressure) as a draw solution for an FO process, which normally requires high osmotic pressure. In this way, the draw solution can be regenerated continuously, without needing to replace it periodically.

The pilot setup is equipped with two 200 L tanks corresponding to the feed solution for the RO process (represented as T2 in Fig. 1) and the feed solution for the FO process (represented as T1 in Fig. 1). To study the permeate flux, the weight of the feed and draw solution tanks was continuously recorded by means of two balances (Kern, Germany) connected to a laptop. The plant contains three pumps (P1 and P2 were self-priming centrifugal pumps, whereas P3 was a high-pressure piston pump). Also, it was equipped with a SW30-2540 spiral-wound membrane (DuPont, USA) as the RO membrane, with an active area of 2.8 m<sup>2</sup>, and a HFFO14® membrane (Aquaporin, Denmark) as the FO membrane,

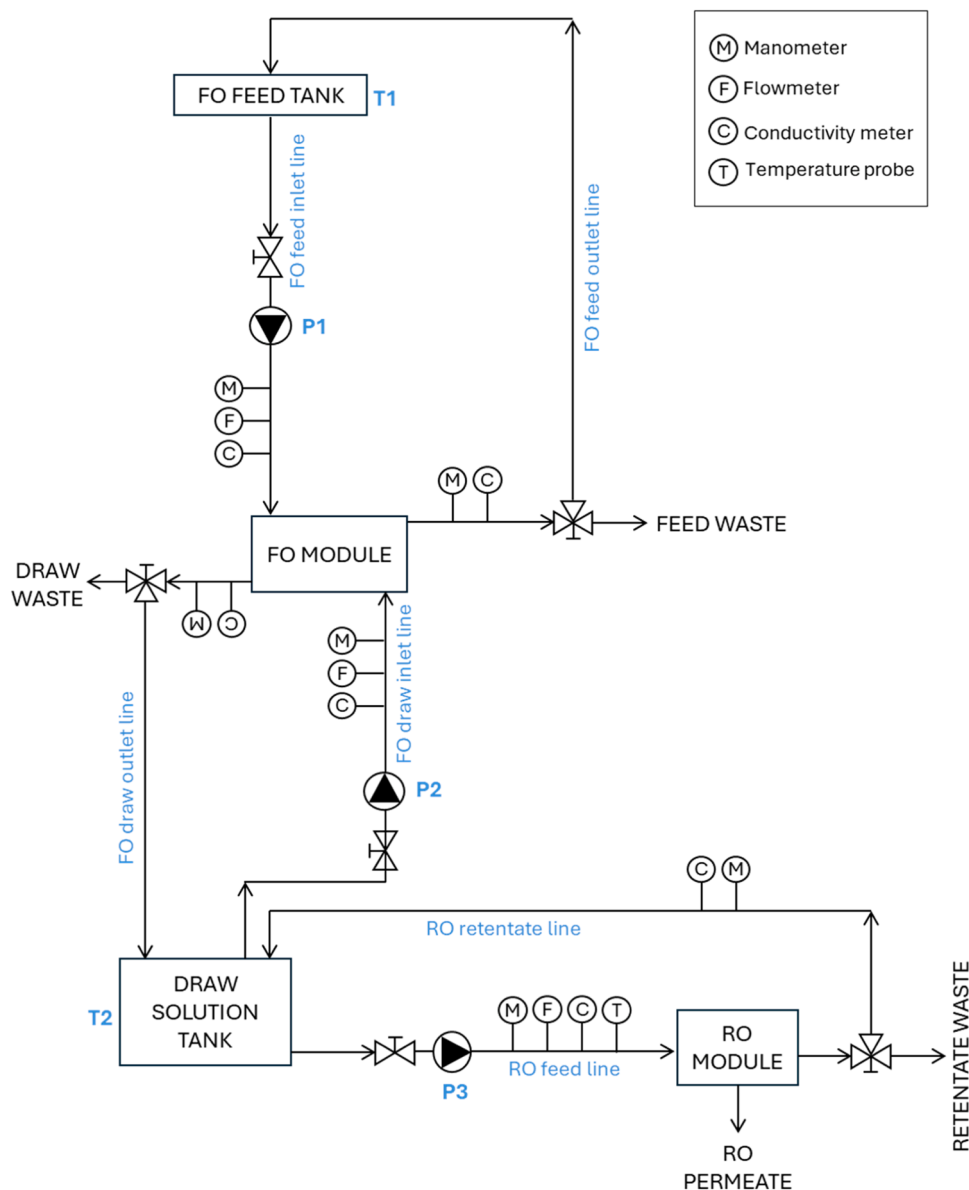


Fig. 1. Diagram of the pilot plant, equipped with the HFFO14® membrane (Aquaporin, Denmark) and the SW30-2540 reverse osmosis membrane (DuPont, USA).

with an active area of 13.8 m<sup>2</sup>. The hollow-fiber Aquaporin membranes have been marked as more efficient than thin-film composite membranes based on cellulose triacetate [20]. A 1 µm cartridge filter was located in the FO line, right before the FO membrane, due to the potential presence of particles or fibers deriving from the textile pieces.

Table 2 contains the characteristics of both membranes. Although both membranes are made of the same material, the pH range of each membrane is different. The pH range is due to a combination of the limitations attributed to the membrane coating, support material, and module material. The same reasoning can be applied to the materials used for the fibers and the module itself. Therefore, internal tests reported by Aquaporin indicated the necessity to limit the pH range to ensure the integrity of the HFFO14® membrane (including a safety margin).

A diagram reflecting the FO and RO sections of the pilot plant is presented in Fig. 1. The FO module was operated in counter-current.

As can be seen in the diagram, the RO retentate was recirculated to the draw solution tank (T2), in order to reestablish the osmotic pressure of the saline solution employed as draw solution.

### 2.3. Operation of the membrane processes

Before proceeding to the FO-RO combined operation, quality control was performed to test the RO and FO lines separately.

#### 2.3.1. Forward osmosis line testing

To assess the quality control of the FO line, the water permeability of the HFFO14® membrane was studied, employing NaCl 0.5 M (with an osmotic pressure of 24.9 bar at 25°C) as the draw solution. The flow rate was set at 400 L/h and 200 L/h for the feed and draw solution inlet, respectively, as recommended by the manufacturer. Similarly, the transmembrane pressure (feed to draw) was set at 0.2 bar. This test lasted 50 minutes.

#### 2.3.2. Reverse osmosis line testing

To assess the quality control of the RO line, a 0.5 M NaCl solution was employed as the feed stream. Three different transmembrane pressures were tested (30, 40 and 50 bar), at a feed flow rate of 650 L/h, during 30 minutes. The conductivity rejection was calculated according to the following formula:

$$R (\%) = 1 - \frac{\text{Conductivity}_p}{\text{Conductivity}_f} \cdot 100 \quad (3)$$

where  $\text{Conductivity}_p$  refers to the conductivity in the RO permeate and  $\text{Conductivity}_f$  refers to the conductivity in the RO feed solution, which corresponds to the FO draw solution (Fig. 1).

**Table 2**  
Characteristics of the reverse osmosis and forward osmosis membranes.

| Parameter                                        | SW30-2540                     | HFFO14®                                                      |
|--------------------------------------------------|-------------------------------|--------------------------------------------------------------|
| Process                                          | Reverse osmosis               | Forward osmosis                                              |
| Brand                                            | DuPont                        | Aquaporin                                                    |
| Material                                         | Polyamide thin film composite | Polyamide thin film composite with integrated water channels |
| Membrane configuration                           | Spiral-wound                  | Hollow-fiber                                                 |
| Membrane area (m <sup>2</sup> )                  | 2.8                           | 13.8                                                         |
| Water flux (L/h·m <sup>2</sup> )                 | -                             | 11.0 ± 1.5 <sup>a</sup>                                      |
| Specific reverse salt flux (g/h·m <sup>2</sup> ) | -                             | 0.15 ± 0.05 <sup>a</sup>                                     |
| Temperature (°C) and pH range                    | < 35 (pH 1-11)                | 5-35 (pH 3-8)                                                |

<sup>a</sup> Testing conditions reported by the manufacturer: 0.5 M NaCl (2.9 %) draw solution vs. deionized water, 25°C, single pass mode, counter-current flow, 400 L/h feed flow rate, 200 L/h draw solution flow rate, 0.2 bar transmembrane pressure (feed to draw).

#### 2.3.3. In-line combination of reverse osmosis and forward osmosis in the pilot plant

After testing the RO and FO lines separately, the in-line combination of both processes was assessed.

First, the flow of each stream was examined. Both the FO and RO processes were intended to occur simultaneously, in a closed loop. The link of both processes was focused on the draw solution tank (see Fig. 1). The outlet of the draw solution was collected in this tank, while the inlet of the draw solution was pumped from this tank to the FO module. At the same time, the RO feed inlet was pumped from the same tank to the RO module, and the RO retentate was collected in the tank, as shown in Fig. 1. Therefore, it was essential to check that both processes were balanced, and the flow rate of each stream was finely adjusted. As a consequence, an excess or an absence of fluid in the draw solution tank was intended to be avoided.

For the RO process, a feed flow of 650 L/h was applied. The transmembrane pressure was set at 50 bar, after studying several pressures, according as described in Section 2.3.2. These operating conditions were maintained throughout the experiment. On the contrary, the flow rates of the feed and draw solution of the FO process were modified to adjust the volume available in the draw solution tank (T2). When the feed flow and draw solution flow specified in Section 2.3.1 (400 L/h and 200 L/h, respectively) were employed during the simultaneous FO-RO process, an excess of volume accumulated in T2, as the difference between the inlet and the outlet volumes was above 120 liters in an hour. Therefore, several combinations of flows for each pump were studied to avoid this situation. The most successful case is reflected in Table 3. In this scenario, the FO feed flow was maintained at 400 L/h, whereas the draw solution flow was set at 80 L/h to minimize the net inflow to. As a consequence, the excess volume in T2 was reduced from 120 liters to 60 liters. To manage this volume excess, a purge valve was installed in T2. The tank was purged every 3 hours to withdraw 0.7 M NaCl to be reused in further stages of the study.

**2.3.3.1. Short-term operation.** In this study, the feed for the FO process was a real end-of-pipe textile wastewater, which was obtained at the outlet of the equalization tank in a textile industry. 0.7 M NaCl was employed as draw solution. This concentration was selected according to a previous work [29], aiming to ensure a sufficient osmotic pressure in the draw solution. The flow rate was set at 400 L/h for the feed stream and at 80 L/h for the draw solution. The transmembrane pressure (feed to draw) was 0.2 bar. Regarding the RO line, the feed flow was adjusted to 650 L/h and the transmembrane pressure was 50 bar. The operation time was 2 hours.

**2.3.3.2. Longer-term operation.** Afterwards, the membrane performance was studied during a longer-term operation and more severe conditions. For this purpose, the concentrated textile wastewater that was obtained from the first set of in-line experiments described in Section 2.3.3.1 was employed as the feed solution of a subsequent FO-RO process. The characteristics of this concentrated textile wastewater are presented in Table 5. Thus, the membrane was submitted to a more challenging feed solution, during a longer period of operation (6 hours). The same operating conditions as in Section 2.3.3.1 were applied. This experiment was repeated twice in order to confirm the reproducibility regarding the membrane performance. The RO step was conducted at a flow rate of 650 L/h and 50 bar.

#### 2.3.4. Cleaning in place

The plant, including the FO and RO lines, was cleaned with deionized water during 15 minutes after each experiment. This flushing was sufficient to clean the RO line. Furthermore, an osmotic backwash was applied during 1 hour to clean the FO membrane. 0.7 M NaCl, whose osmotic pressure is 34.7 bar at 25°C, was used as the feed solution. Deionized water was used on the other side of the membrane.

**Table 3**

Flow (L/h) of each stream involved in the simultaneous FO-RO process. FO feed flow was 400 L/h and draw solution flow was 80 L/h. RO was performed at 650 L/h and 50 bar.

| Streams entering the DS tank |                    |                   |                                  | Streams exiting the DS tank |                     |                                   | Result (L/h) |
|------------------------------|--------------------|-------------------|----------------------------------|-----------------------------|---------------------|-----------------------------------|--------------|
| RO Retentate (L/h)           | FO DS Outlet (L/h) | FO permeate (L/h) | Total inlet in the DS tank (L/h) | FO DS inlet (L/h)           | RO feed inlet (L/h) | Total outlet of the DS tank (L/h) |              |
| 600                          | 80                 | 110               | 790                              | 80                          | 650                 | 730                               | 60           |

Afterwards, the water permeability of the membrane was assessed, employing 0.5 M NaCl, whose osmotic pressure is 24.8 bar at 25°C, as draw solution.

## 2.4. Analysis of the streams

The characterization of the textile wastewater involved several analytical methods. COD was determined using the Dichromate methodology as described by Bendicho and Lavilla [30]. A Hach Lange DR3900 spectrophotometer (Hach Lange, USA) was utilized for this purpose. Total organic carbon (TOC) analysis was conducted using a multi N/C 3100 instrument (Analytik Jena, Germany). pH measurements were performed using a Jenway 3510 pH meter. TDS and conductivity were measured with a ULTRAMETER II™ 4P (Myron L®, USA). Additionally, the osmolality of the samples was determined using an Osmomat 030 (Gonotec, Germany). Osmolality indicates the total concentration of osmotic-active particles in the sample without requiring knowledge of their individual properties or dissociation characteristics. Thus, the determination of osmolality was preferred over the estimation of the osmotic pressure (applying the Van't Hoff equation or even other equations developed to estimate the osmotic pressure [31]), because the exact composition of the textile wastewater was unknown and the estimation of the dissociation of each compound implied a high error.

## 2.5. Development of a preliminary input analysis

The energy ( $E$ ) requirements of the simultaneous FO-RO process presented in this work were calculated considering the power employed by each pump included in the plant. The following equation was used:

$$E = \frac{\text{Power}}{Q_w} = \frac{Q_s \cdot \text{Pressure}_{\text{pump}}}{J_w \cdot A} \quad (4)$$

where  $Q_s$  is referred to the flow rate of each stream (feed solution, draw solution or RO feed stream). The pressure and flow rate provided by each pump (as reported in Section 2.3.3.1) were considered.  $Q_w$  and  $J_w$  represent the permeate flow rate and permeate flux of clean water obtained as the RO permeate and  $A$  represents the RO membrane area. The efficiency ( $\eta$ ) was considered to be 85% for all pumps.

## 3. Results and discussion

### 3.1. Composition of the textile wastewater

Table 1 (in Methodology section) contains the characterization of the textile wastewater used as feed stream. The TDS content was higher than other textile effluents reported in literature [20,32]. Among the compounds present in end-of-pipe wastewater, a mixture of dyes, defoaming components and emulsifiers, salts, surfactants and other auxiliary products can be found, all of them contributing to the TDS load of the sample.

### 3.2. Individual testing of the forward osmosis and reverse osmosis lines

Before proceeding with the simultaneous operation of the FO and RO units, both lines were studied independently, in order to ensure the

correct performance of both membranes, as a quality control. To that end, deionized water was employed as feed for the FO membrane, whereas 0.5 M NaCl was used as feed solution for the RO membrane. The results, regarding permeate flux, reverse flux salt flux, and conductivity rejection obtained with the employed membranes are reflected in Fig. 2.

The results obtained for the HFFO14® membrane (Fig. 2A) were in accordance with the previous literature [33] and considered satisfactory, as reverse salt flux ( $J_s$ ) was very low ( $0.79 \pm 0.02$  g/h·m<sup>2</sup>) in comparison with water flux. In the case of the SW30-2540 membrane (Fig. 2B), high rejections of conductivity were obtained at all tested pressures (30, 40, and 50 bar). Being NaCl the only ionic species present in the solution, high rejection of NaCl was attributed to the SW30-2540 membrane [34,35]. These high rejections were aimed, as the objective of the RO was to concentrate a saline solution to be used as draw solution for the FO process when FO and RO work simultaneously in the membrane plant. A slight increase in rejection as pressure rises can be observed, as expected [36]. Regarding the permeate flux obtained with the SW30-2540 membrane, a linear relation with the transmembrane pressure was observed, indicating low fouling potential. The recovery reached 13.5% at 50 bar. Therefore, to increase the productivity of the process, 50 bar was selected as the transmembrane pressure for the following experiments.

Once both membranes were tested individually, the combined FO-RO operation was assessed.

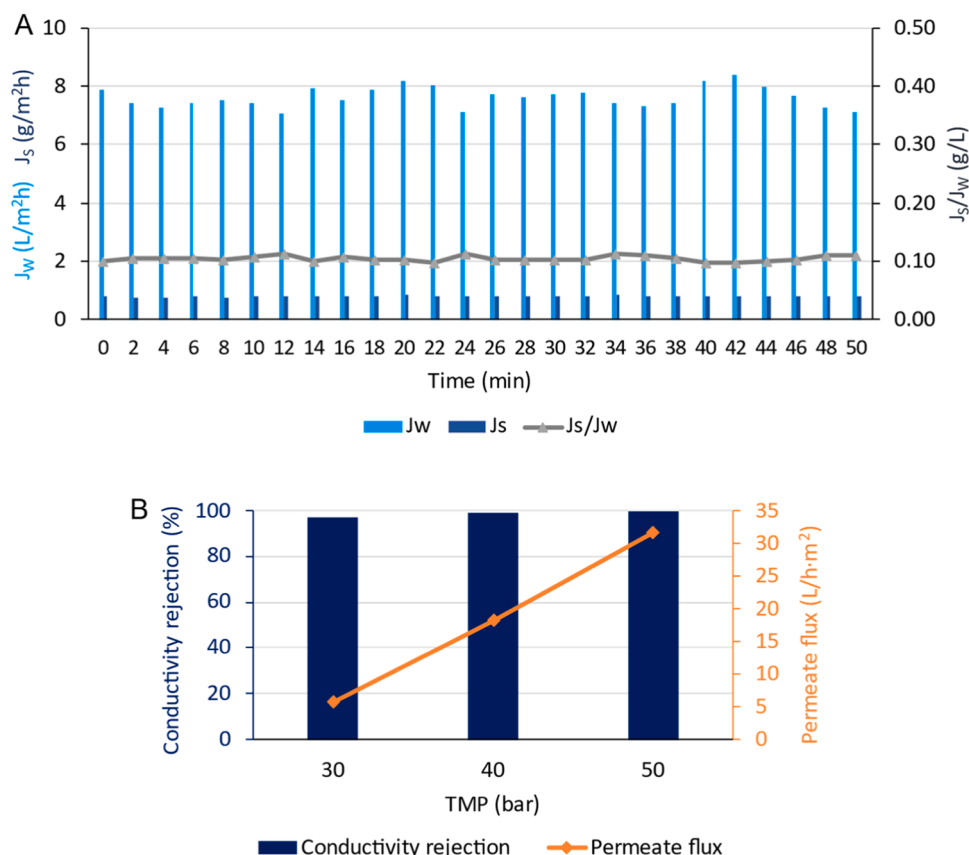
### 3.3. In-line combination of forward osmosis and reverse osmosis

#### 3.3.1. Short-term operation

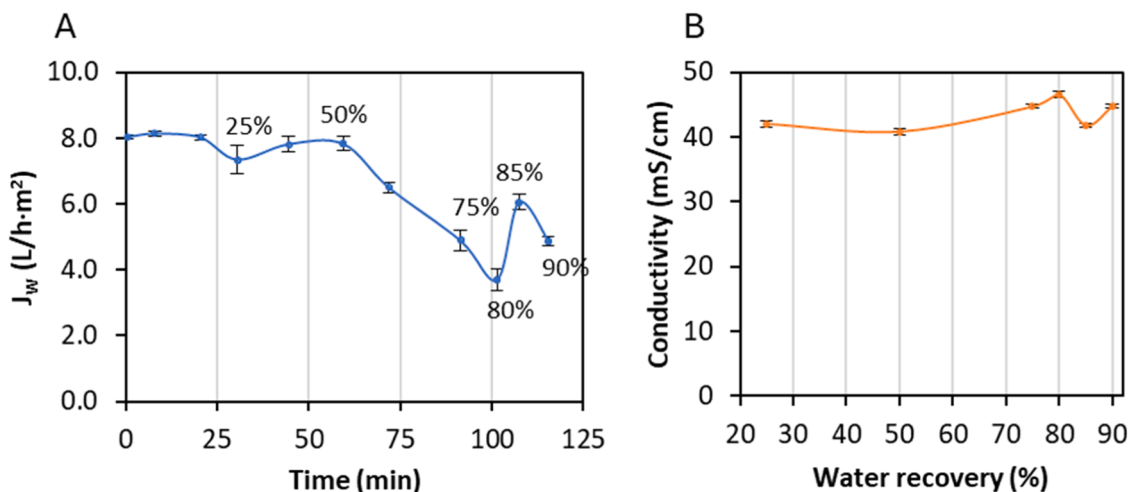
When both membrane modules (RO and FO) were working at the same time, a continuous regeneration of the draw solution was performed. Thus, as the conductivity of the draw solution remained high during the whole run (Fig. 3B), a constant permeate flux was registered during the FO step, as reflected in Fig. 3A. This is the first time that the permeate flux obtained during the treatment of real wastewater with the HFFO14® membrane is reported in the scientific literature.

According to Fig. 3A, the permeate flux obtained with the HFFO14® membrane was constant during the first hour of the process, which corresponded with a 50% recovery of water from the end-of-pipe textile wastewater. Afterwards, the permeate flux decreased. However, the conductivity of the draw solution did not decrease (Fig. 3B). This fact indicated that the draw solution did not become diluted, which was the main objective of this study. Hence, the osmotic strength of the draw solution was not altered as a result of water crossing the FO membrane. The reduction in permeate flux that was observed when the recovery was greater than 50% can be attributed to the progressive fouling of the membrane as the feed solution became more concentrated. Also, the saturation of the cartridge filter installed in the plant was observed at this point and contributed to the permeate flux decline. Due to this saturation, the cartridge filter was replaced when 80% water recovery was achieved. Between 60 and 100 minutes of operation, an increase in the conductivity of the draw solution was observed (Fig. 3B). During this period of time, the draw solution was getting concentrated by the RO process, however, the corresponding dilution by the FO water permeate was reduced, because  $J_w$  was lower.

After replacing the cartridge filter, the observed FO permeate flux increased, as observed for a 85% recovery. However, after 107 minutes



**Fig. 2.** A: Permeate flux ( $J_w$ ) and reverse salt flux ( $J_s$ ) obtained with the HFFO14® membrane, employing deionized water as feed solution (400 L/h) and 0.5 M NaCl as draw solution (200 L/h). B: rejection of NaCl and steady-state permeate flux obtained with the SW30-2540 membrane at 650 L/h and 30, 40, and 50 bar, during the filtration of 0.5 M NaCl.



**Fig. 3.** Permeate flux ( $J_w$ ) (A) and conductivity of the draw solution (B) during the treatment of the textile wastewater with the HFFO14® membrane. The draw solution (0.7 M NaCl) was regenerated continuously by coupling the process with a reverse osmosis process, employing the SW30-2540 membrane. The percentage of water recovery has been annotated in panel A.

of FO operation and 85% water recovery, the concentration of solutes in the feed solution was much higher than their concentration in the raw wastewater. Therefore, the cartridge filter became saturated again, which, together with the membrane fouling, prompted a progressive decline of the permeate flux until 90% of the water was recovered.

The proposed combination of the FO and RO loops allowed the development of a highly efficient process. By regenerating the draw

solution, 90% of water recovery was achieved, which reflects an improvement of previous results obtained without the regeneration of the draw solution. For example, Mogashane et al. reported a water recovery of 71% when concentrating a brine containing 20 g/L Na<sub>2</sub>SO<sub>4</sub> with the same membrane, employing 80 g/L NaCl as the draw solution [37]. Also, Sbardella et al. employed the HFFO14® membrane, reaching a 70% water recovery when concentrating landfill leachate, using a 100

mS/cm draw solution [38].

The regeneration of the draw solution ensured the maintenance of its osmotic pressure during the whole FO process, which was highly effective. Furthermore, the application of a simultaneous RO to conduct the regeneration allowed for the improvement of the sustainability of the process. The preparation of a new synthetic draw solution to restore the driving force would be costly and time-consuming; however, the RO process ensures the recycling of the same draw solution with a much lower time and energy consumption.

As a result of the achieved water recovery, the textile wastewater was concentrated. Fig. 4 illustrates the visual concentration of the textile wastewater during the progression of the FO process, as water recovery percentage increased. Even though the main advantage of the simultaneous FO-RO process is the obtainment of clean water from waste effluent, the concentrated textile wastewater could also be considered a source of valuable chemicals. According to Shu et al., textile wastewater may contain up to 40% of functional dyes that were not fixed to the fabric during the dyeing process [39]. Therefore, the concentrated stream obtained during the FO could be employed as the starting point for a new dyebath, reducing the use of resources.

The increase in concentration that is perceptible in Fig. 4 is also supported by the high rejection values that are presented in Table 4, which shows the rejection of COD and TOC obtained with the HFFO14® membrane. In all cases, the rejection of COD and TOC was over 99%. Normally, the rejection of COD and TOC is aligned, as both are measurements related to the organic matter present in the sample [40,41]. These data show a high retention capacity of the membrane. Furthermore, the absorbance spectrum of the draw solution after the short-term experiment is presented in Fig. 5. This indicated that the draw solution was not colored, as dyes from the textile wastewater did not cross the membrane.

The values presented in Table 4 and Fig. 5 indicate that water molecules were practically the only compounds that crossed the membrane. Therefore, the proposed process, employing the HFFO14® membrane, enabled the obtainment of clean water from textile wastewater.

Regarding the reverse salt flux, it was evaluated during the processing of deionized water (Fig. 2). However, the evaluation of this parameter during the processing of a real stream is not straightforward, because the feed solution contains so many compounds that the detected conductivity cannot be solely attributed to the presence of sodium or chloride ions. Furthermore, the individual determination of these ions (e.g. using commercial kits or specific probes) becomes nonspecific in such a complex matrix, leading to analytical errors. Nevertheless, in this particular case, a reverse flux of salts to the feed solution was not considered problematic for the textile industry. This is because salts are required during the dyeing stage to fix the color to the fabric. Therefore, if the concentrated textile wastewater were to be reused in a subsequent dyeing step, the presence of salt could represent a benefit for the process, as these ions would be already present in the solution [42].

Table 5 contains the characterization of the textile wastewater after the FO process. A high increase in the osmolality and osmotic pressure of the effluent was observed, in comparison with the values reported in Table 1. Similarly, conductivity, COD, and the concentration of total dissolved solids raised up, as a result of the high withdrawal of water from the textile wastewater.

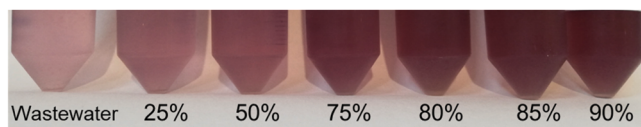


Fig. 4. Concentration of the textile wastewater as water recovery percentage increased during the forward osmosis process.

Table 4

Rejection of chemical oxygen demand (COD) and total organic carbon (TOC) obtained with the HFFO14® membrane during the treatment of textile wastewater in the in-line combination of forward osmosis and reverse osmosis. The employed draw solution was 0.7 M NaCl. Relative standard deviation of the data was below 12%.

| Water recovery (%) | Rejection of COD (%) | Rejection of TOC (%) |
|--------------------|----------------------|----------------------|
| 25                 | -                    | 99.3                 |
| 50                 | 100.0                | 99.7                 |
| 75                 | 98.7                 | 99.1                 |
| 90                 | 100.0                | 99.9                 |

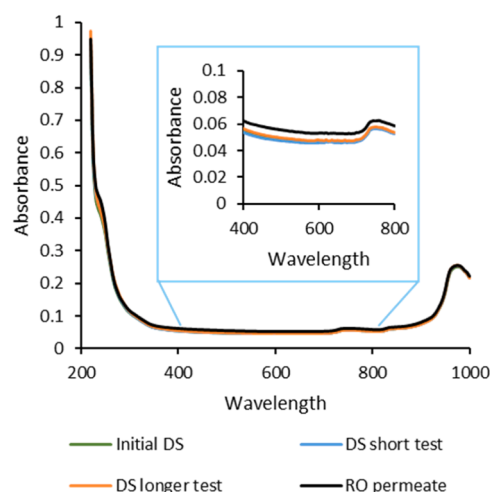


Fig. 5. Absorbance spectra of the draw solution (DS) and reverse osmosis (RO) permeate.

Table 5

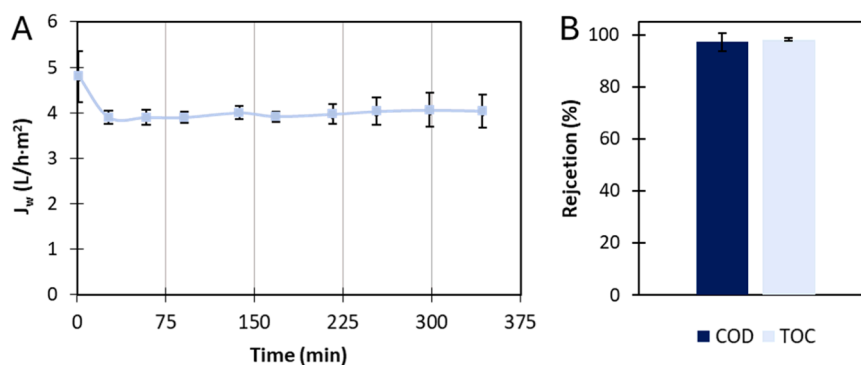
Characterization of the textile wastewater before and after its concentration (90% water recovery) by forward osmosis in the in-line combination of forward osmosis and reverse osmosis.

| Parameter                                | End-of-pipe wastewater (from Table 1) | Concentrated end-of-pipe wastewater |
|------------------------------------------|---------------------------------------|-------------------------------------|
| COD (mg/L)                               | 140                                   | 2130                                |
| Total dissolved solids (mg/L)            | 2583                                  | 20835                               |
| Conductivity ( $\mu\text{S}/\text{cm}$ ) | 3394                                  | 21725                               |
| Osmolality (osmol/kg)                    | 0.05                                  | 0.4                                 |
| pH                                       | 7.3                                   | 7.7                                 |

### 3.3.2. Longer-term operation

As explained in Section 2.3.3.2, this concentrated end-of-pipe wastewater (Table 5) was again employed as the feed solution in a subsequent treatment in the pilot plant. This strategy allowed us to test the performance of the FO membrane under more demanding conditions. As can be seen in Fig. 6A, the permeate flux suffered an initial drop, as a result of the membrane fouling. Fouling of the membrane was expected in this experiment, considering that the feed solution was much more concentrated than in the previous experiment, whose results are shown in Fig. 3.

After the first 30 minutes, when the process was stabilized and external concentration polarization occurred, a stable permeate flux was observed. This initial decline was not present during the short-term operation (Fig. 3A) because a much lower concentration of solutes (Table 5), which prevented their rapid accumulation on the membrane surface [43]. As shown in Fig. 6A, the permeate flux varied between 3.9 and 4.1  $\text{L}/\text{h}\cdot\text{m}^2$ . In this case, the permeate flux was lower than the



**Fig. 6.** Permeate flux (A) and FO rejection values (B) obtained during the treatment of the previously concentrated end-of-pipe wastewater, employing the HFFO14® membrane. The volume concentration factor was 1.65. The draw solution (0.7 M NaCl) was regenerated continuously by coupling the process with a reverse osmosis process, employing the SW30-2540 membrane. Error bars indicate the standard deviation between the replicates.

permeate flux observed during the short-term operation. This was because the feed solution was much more concentrated. In fact, the concentration of TDS in the feed was eight times higher in the longer-term operation, whereas the same draw solution (0.7 M NaCl, corresponding to 1.4 osmol/kg) was employed. Thus, the draw osmolality/feed osmolality ratio was 3.5, whereas this factor was 26.9 during the short-term experiment. This indicated that the driving force of this process was lower, supporting the observed lower permeate flux.

Nevertheless, a stable permeate flux was obtained with the HFFO14® membrane. Avoiding the progressive decay of the flux constitutes an improvement in relation to the current state-of-the-art. For example, in the study of Korenak et al., 50% of water flux decrease was observed during the concentration of textile wastewater, due to the dilution of the draw solution [20]. These results underline the effectiveness of regenerating the draw solution, as its osmotic strength stayed constant and did not decrease within the longer-term operation, allowing the maintenance of a stable permeate flux during the six hours of the process.

Several researchers tested RO to treat wastewater from the textile industry, but, due to severe membrane fouling, a pretreatment (ultrafiltration (UF), membrane bioreactors (MBR), coagulation-flocculation, resin adsorption, etc.) was necessary before the RO process [44,45]. Liu et al. collected information of current applications of membrane technology to treat wastewater from the dyeing industry in China [46]. They concluded that RO has become the dominant membrane technology process (more than 80% of the applications in terms of both capacity and number of installations), because of water reuse incentives and water quality standards. However, RO was always used combined with some pretreatment process, such as UF and MRB. In this work, we demonstrate that when FO is used to concentrate the textile wastewater, the pretreatment can be avoided, which may reduce both capital expenditures (CAPEX) and operational expenditures (OPEX), since no additional equipment or chemicals (to conduct a pretreatment) are needed and operational complexity is decreased [47]. The HFFO14 membrane maintained stable permeate flux even with untreated textile wastewater, indicating that fouling was not significant under the tested conditions. Therefore, combining FO and RO can be more energetically and economically beneficial than using RO alone, due to the reduced membrane fouling and associated maintenance requirements. Because of the high transmembrane pressures required, fouling is a common issue in RO that reduces efficiency and requires frequent cleaning, especially if fouling-prone streams, such as textile wastewater, are treated. However, the concentration of NaCl solutions by RO does not result in significant membrane fouling, as observed in this work.

Also, the stable permeate flux reflected in Fig. 6A indicates that the simultaneous FO-RO can be operated for a longer time, if a higher concentration of the textile wastewater is aimed. The time evolution of the operating conditions (feed and draw temperatures, and RO flow and

TMP) is presented in Supplementary Figure 1. The RO line displayed excellent stability, ensuring the long-term functioning of the system. Regarding the temperature of the FO streams, an increase was observed throughout the time, due to the effect of the high-pressure piston pump (P3, in Fig. 1). The industrial application of this technology is oriented towards the treatment of the textile wastewater and, additionally, the recovery of clean water. In this scenario, the increment of the temperature (always inside the membranes temperature range) would not be very detrimental for the operation, because the quality of the streams would not be compromised by a higher temperature. Nevertheless, when necessary, a heat exchanger can be integrated into the setup to regulate feed temperature. Thus, the achievement of longer operation times, without affecting the membrane performance, contributes to the spacing of the cleaning steps, then increasing the sustainability of the process. The volume reduction factor reported in Fig. 6 only refers to the longer-term operation, during the 6 hours of experiment. However, if the global concentration of the textile stream is considered, from the initial textile wastewater that was collected to the final concentrated stream after the longer-term operation, a concentration factor of 16.5 was achieved. This implies a 93.9% water recovery. However, when RO was used independently, the highest recovery rates achieved in the literature were around 70% [46]. In this regard, Wei et al. [44] concluded that the recovery rates should not exceed 60%.

Fig. 6B contains the rejection of COD and TOC that was obtained with the HFFO14® membrane when the concentrated end-of-pipe wastewater was treated. In all cases, these values were above 92%, which ensured an adequate retention of the compounds.

Apart from reducing the volume of the end-of-pipe textile wastewater, the proposed process allowed us to obtain clean water as the RO permeate. The permeate flux obtained with the SW30-2540 membrane was stable at  $22.2 \pm 0.4$  L/h·m². Table 6 contains the characterization of the obtained permeate and retentate from the RO loop.

As can be seen in Table 6, the characteristics of the RO retentate were maintained throughout the FO-RO process. Only a slight dilution

**Table 6**

Characterization of the reverse osmosis retentate and permeate obtained with the SW30-2540 (650 L/h, 50 bar) during the combined forward osmosis and reverse osmosis treatment of textile wastewater.

| Parameter                     | Retentate (beginning of the process) | Retentate (end of the process) | Permeate |
|-------------------------------|--------------------------------------|--------------------------------|----------|
| COD (mg/L)                    | 82.5                                 | 73.5                           | 3.2      |
| Total dissolved solids (mg/L) | 70240                                | 64420                          | 389.1    |
| Conductivity (mS/cm)          | 59.1                                 | 55.7                           | 0.56     |
| Osmolality (osmol/kg)         | 1.2                                  | 1.12                           | 0.02     |
| pH                            | 7.2                                  | 7.4                            | 6.4      |

occurred (resulting in the mild reduction of COD, total dissolved solids, and osmolality), but no significant differences were observed ( $p < 0.05$ , according to the two-sample t-test performed) between the conductivity of the initial saline solution and the final draw solution. These results confirmed that, even though the RO membrane rejected most of the solutes, the flow rates were correctly balanced. Therefore, the continuous concentration of the RO retentate was compensated with the continuous inlet of the draw solution. Thus, the continuous regeneration of the draw solution was considered successful. On the contrary, the permeate was a stream extremely reduced in organic matter, as reflected by its low concentration of COD. The conductivity and total dissolved solids were also highly eliminated. This characterization, together with the pH and osmolality of the permeate, indicated that clean water was obtained after treating the textile wastewater by the proposed simultaneous FO-RO process. This clean water can be effectively reused in various industrial applications, including the textile sector. Due to its low salinity and minimal organic load, this water is well-suited for reuse in dyeing baths, fabric rinsing, and equipment cleaning processes, where consistent water quality is essential. The treatment of textile wastewater and subsequent reuse of RO permeate proposed in this study can reduce freshwater consumption and also minimize wastewater discharge, enabling more sustainable and cost-efficient production cycles in water-intensive industries.

### 3.4. Cleaning of the membrane

After the treatment of the end-of-pipe textile wastewater, the FO membrane was cleaned in order to be reused. Fig. 7 contains the results regarding the water permeability of the membrane after its use with the textile wastewater and after the cleaning process.

Considering the results observed in Fig. 6A and the characteristics of the feed solution, the deposition of solutes on the membrane surface is suggested. After performing the longer-term test, the plant was rinsed with deionized water, and the water permeability of the FO and RO membranes was immediately evaluated. In the case of the RO membrane, the water permeability was  $1.2 \text{ L/h}\cdot\text{m}^2\cdot\text{bar}$ , which represented a 100% recovery of the initial water permeability of the SW30-2540 membrane. The permeability of the FO membrane, as shown in Fig. 7, was slightly lower than the initial permeability. In fact, the water permeability of the HFFO14® membrane after the longer-term test and the water rinsing corresponded to 95% of the initial permeability. This indicated that the simple rinsing of the plant was sufficient to remove most of the foulants from the membrane surface and, therefore, the fouling of the membrane was not severe. Afterwards, an osmotic backwash was performed, which allowed the complete recovery of the initial permeability of the membrane.

As indicated in Section 2.1, the textile wastewater treated in this work was derived from an industry employing reactive dyes during the dyeing stage. Reactive dyes are designed to interact with cellulosic fibers, establishing covalent bonds with the hydroxyl (-OH) groups of the

polymer [48]. However, the FO membrane employed in this study is not made of cellulose. As reported in Table 2, the material of the HFFO14® membrane is polyamide. Polyamide contains amide (-CONH-) groups in the main chain of the polymer and amine (-NH<sub>2</sub>) and (-COOH) groups in terminal positions. Even though these functional groups can interact with some dyes, the reactive dyes are not designed to establish covalent bonds with polyamide. Therefore, the affinity between the membrane and the reactive dyes present in the end-of-pipe wastewater was expected to be low. Also, it is important to consider that end-of-pipe wastewater is quite diluted, as it is a mixture of all the effluents generated in the industry. As commented previously, the end-of-pipe wastewater is composed of discarded dyebaths, but also aqueous streams from rinsing, pre-washing and post-washing steps [3]. Therefore, this wastewater contains diluted dyes whose chemical structure does not present a high affinity for the membrane active layer. Finally, as FO operates without the application of significant external pressure, the fouling layer is not highly compacted, which facilitates its elimination. These results are in line with previous works dealing with the cleaning of FO membranes [32].

Regarding the reverse salt flux, it increased after the treatment of the end-of-pipe wastewater, in comparison with the values obtained with the pristine membrane. This increment was attributed to the stress suffered by the HFFO14® membrane during the longer-term operation. As the FO process progresses, more salts can penetrate the membrane from the draw solution side, generating a concentration polarization of salt inside the membrane structure. In this scenario, the salt concentration at the membrane-feed solution interface increases, which can lead to an increment of the reverse salt flux [49,50]. Also, small injuries on the membrane active side can be suggested, as the reverse salt flux did not decrease after the backwash. Nevertheless, the specific reverse flux ( $J_s/J_w$ ) was satisfactory, as it was not higher than  $0.18 \text{ g/L}$ , indicating that the integrity of the membrane was not compromised and could be easily cleaned and reused. Also, the preservation of the integrity of the membrane was confirmed after replicating the test presented in Fig. 6, when the concentrated textile wastewater was treated. The membrane performance was very similar for the two replicates, indicating that it did not lose its selectivity properties.

The recovery of the hydraulic permeability of both the FO and RO membranes after water rinsing not only demonstrates the low fouling tendency of the membranes, but it also implies that the costs associated with membrane cleaning are significantly lower and the overall process is more environmentally friendly compared to conventional RO, where, according to the literature [45,46], chemical cleaning is required.

### 3.5. Preliminary energy input analysis

Combining FO and RO can be more energetically beneficial than using RO alone for treating textile wastewater due to the reduced membrane fouling of FO membranes. Because of the high trans-membrane pressures required, fouling is a common issue in RO that reduces efficiency and requires frequent cleaning, especially if fouling-prone streams, such as textile wastewater, are treated. However, the concentration of NaCl solutions by RO does not result in significant membrane fouling, as observed in this work.

Previous studies involving RO treatment of textile industry wastewater have consistently reported severe membrane fouling, necessitating extensive pretreatment of the feed water, an intervention that was not required in the present work. For instance, Yang et al. indicated a high contribution to RO fouling of microbial products and protein derivatives during the treatment of textile dyeing secondary effluent. To mitigate this fouling during RO, they compared the effect of UF and exchange resin as pretreatments [51]. Also, in the study by Wei et al. [44], the feed water was treated via a membrane bioreactor prior to RO application. Even after applying this pretreatment, all tested RO membranes exhibited fouling, along with a significant decline in permeate flux at recovery rates exceeding 50%. This reduction was attributed to

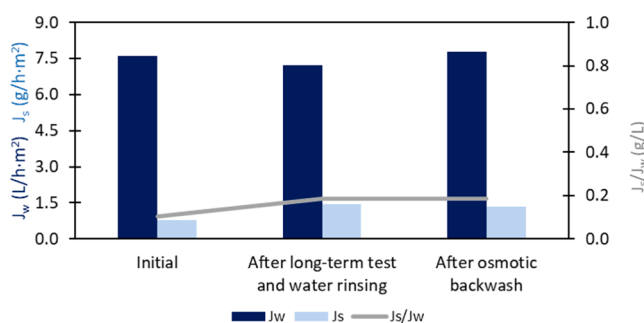


Fig. 7. Water permeability of the HFFO14® membrane after the processing of the end-of-pipe textile wastewater and after its cleaning. Draw solution consisted of 0.5 M NaCl.

the increased osmotic pressure resulting from solute concentration. Consequently, the authors of the study concluded that recovery rates should not exceed 60%. Yin et al. [45] treated the secondary effluent from a textile factory by means of RO. Due to severe membrane fouling, they investigated different pretreatment technologies, including coagulation, ultrafiltration, magnetic ion exchange resin adsorption (MIEX) and their integrations. Despite the pretreatment, fouling was observed on the RO membrane tested by the authors. In contrast, in our work, no pretreatment was considered, and fouling was very low. Liu et al. [46] examined the status of RO in China for the treatment of textile industry wastewater, based on both a review of the scientific literature and real or prospective applications at industrial scale. In many cases, RO was applied after a secondary treatment of the wastewater. In all instances, it was found that pretreatment prior to RO was necessary, with UF being the most commonly used method, followed by membrane bioreactors. The highest recovery rates achieved were around 70%. However, in our work, this pretreatment was not necessary, as the hybrid FO-RO process was not affected by fouling in such a high level.

A preliminary energy input analysis has been performed from the results obtained in this work. For this purpose, the power consumption was estimated for the three pumps included in the pilot plant (Fig. 1), taking into account the flow rate and pressure provided by each pump (Section 2.3.3.1). The efficiency was considered to be 85% for the three pumps. Then, the energy consumed per cubic meter of clean water produced in the RO step was also estimated and presented in Table 7. For the calculations, the overall water recovery for the integrated FO-RO process was 40%, but, as the feed wastewater had already been previously concentrated in the short term test, the overall concentration factor reached was 16.5.

As can be observed, the energy consumed in the FO process is negligible in comparison with the energy consumed in the RO step if the wastewater was treated by RO alone. In that case, the energy consumption would be high due to the reduced permeate flux expected as a consequence of fouling, according to the literature [44–46,51]. Therefore, a pretreatment stage prior to the reverse osmosis process would be necessary, which entails both increased capital investment and higher energy consumption. In the estimations presented in Table 7, no energy recovery from the RO retentate has been considered. As it is well known, different devices, such as turbines, turbochargers, and pressure exchangers, can be used to recover energy from the RO retentate. If some of these devices were included in the plant, energy consumption could significantly decrease. For instance, if the efficiency of the energy recovery system was 98%, which is typical for isobaric devices [52], energy consumption in the RO step would drop to about 4.3 kWh per cubic meter of produced RO permeate. Nevertheless, the RO system could be further optimized to try to reduce as much as possible the energy consumption.

The independent RO process was not tested in this work to treat the wastewater. Nevertheless, regarding energy consumption, it is important to highlight that the integrated FO-RO process was not optimized in the present study, as the primary objective was to demonstrate the feasibility of simultaneous operation of both processes. Nonetheless, by optimizing the feed flow rates of the FO and RO units, as well as the operating pressure of the RO system, it would be possible to achieve substantially lower energy consumption. Unfortunately, the studies that

conducted economic analysis for the treatment of textile wastewater with membrane processes are very scarce and mostly RO as a secondary treatment of the wastewater. Therefore, a direct comparison with the present work is not possible. However, several studies can be cited that compare the integration of FO-RO with RO for the treatment of wastewater from different industries. In many of these studies, seawater was used as the draw solution for FO. For example, Choi et al. [53] demonstrated, using municipal wastewater secondary effluent as feed, that the fouling potential of the RO membrane was greatly reduced by the application of FO. Therefore, the authors reported that the sustainable operation of FO-RO is a realistic possibility because FO acted as an effective pretreatment for the RO membrane. The same authors [54] evaluated a FO-RO pilot plant for treating wastewater from a coal-fired power plant integrated with seawater desalination. Energy consumption in both FO and RO was systematically evaluated during 5 months of pilot operation. The authors claimed that the integrated FO-RO process was able to significantly reduce the RO fouling potential through FO treatment of wastewater. The study also reported energy consumption data and showed that total energy consumption for the integrated FO-RO process was 15% less than that of typical seawater desalination by RO. These results demonstrated that sustainable FO-RO operation can be achieved in the treatment of wastewater owing to easy fouling control, low energy consumption, and superior final water quality.

In this context, Yalamanchili and co-workers [55] performed a design and simulation study on FO-RO hybrid processes for wastewater recovery by FO and seawater (SW) desalination by RO. They reported that the optimal configuration was able to achieve both 90 % wastewater recovery and specific energy consumption below 1 kWh/m<sup>3</sup>. They compared the hybrid process with independent RO schemes where 80% of the total permeate was produced from water reuse, while the remaining 20% came from SWRO desalination. From an energy perspective, independent schemes yielded a 14.5% improvement in energy consumption relative to the FO-RO configuration. However, fouling was not considered in the simulation. Thus, if MF/UF pre-treatment is incorporated as pretreatment prior to the independent RO processes an extra specific energy input of 0.2 kWh/m<sup>3</sup> would be required [56]. In this situation the energy consumption for the integrated FO-RO process would be lower than that for the independent RO processes. Moreover, Yalamanchili et al. [55] reported other benefits associated with the integrated scheme, such as double dense membrane barrier to assure safe water reuse.

Thus, the preliminary energy analysis presented here demonstrates that continuous draw solution regeneration can be achieved with moderate energy input, avoiding the repeated preparation of fresh solutions. There are scientific evidences which emphasizes cost competitiveness of FO at a conceptual level [20], and, moreover, our pilot-scale assessment confirms that the process can also be implemented with energy demands compatible with large-scale application.

#### 4. Conclusions

The proposed combination of forward osmosis and reverse osmosis has been demonstrated to be a suitable strategy for treating textile wastewater while the driving force of the forward osmosis process is maintained. This is supported by the stable values of permeate flux that were obtained with the HFFO14® forward osmosis membrane and the high rejection of chemical oxygen demand and total organic carbon that were observed. Also, the combination of forward and reverse osmosis allowed to obtain clean water as the reverse osmosis permeate. This water can be employed again in the textile industry, for new dying processes and other operations, contributing to the circular economy in this industry.

#### Funding

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**Table 7**

Preliminary estimation of the energy consumed by the pumps in the proposed process (FO permeate flux: 4 L·h<sup>-1</sup>·m<sup>-2</sup>; RO permeate flux: 22 L·h<sup>-1</sup>·m<sup>-2</sup>; RO transmembrane pressure: 55 bar; overall FO-RO recovery: 39.4%).

| Pump    | Power consumption (W) | Energy consumption (kWh/m <sup>3</sup> RO permeate) |
|---------|-----------------------|-----------------------------------------------------|
| Feed FO | 4.9                   | 0.04                                                |
| Draw FO | 0.4                   | 0.006                                               |
| RO      | 1062                  | 17.24                                               |

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### CRedit authorship contribution statement

**Carmen M. Sánchez-Arévalo:** Writing – original draft, Visualization, Validation, Investigation, Data curation. **Laura García-Suarez:** Validation, Methodology. **Maria Salud Camilleri-Rumbau:** Writing – review & editing, Visualization, Supervision, Investigation, Conceptualization. **Jörg Vogel:** Writing – review & editing, Validation, Resources, Investigation. **Silvia Álvarez-Blanco:** Writing – review & editing, Visualization, Validation, Project administration, Investigation. **M. Cinta Vincent-Vela:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization. **Beatriz Cuartas-Urbe:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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### Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.cep.2025.110689](https://doi.org/10.1016/j.cep.2025.110689).

### Data availability

<https://zenodo.org/communities/textmem/> (The data corresponding to this study will be openly published in the Zenodo community of the TEXTMEM research project () upon publication of the article.)

### References

- [1] N. Abidi, E. Errais, J. Duplay, A. Berez, A. Jrad, G. Schäfer, M. Ghazi, K. Semhi, M. Trabelsi-Ayadi, Treatment of dye-containing effluent by natural clay, *J. Clean. Prod.* 86 (2015) 432–440, <https://doi.org/10.1016/j.jclepro.2014.08.043>.
- [2] T. Hussain, Textile Processing, *Text. Eng.* 5 (2016) 83–110, <https://doi.org/10.1515/9783110413267-007/HTML>.
- [3] D.A. Yaseen, M. Scholz, Textile dye wastewater characteristics and constituents of synthetic effluents: a critical review, *Int. J. Environ. Sci. Technol.* 16 (2019) 1193–1226, <https://doi.org/10.1007/s13762-018-2130-z>.
- [4] J. Sharma, S. Sharma, V. Soni, Classification and impact of synthetic textile dyes on aquatic flora: a review, *Reg. Stud. Mar. Sci.* 45 (2021) 101802, <https://doi.org/10.1016/J.RSMA.2021.101802>.
- [5] M. Tabish, A.B. Tabinda, Z. Mazhar, A. Yasar, J. Ansar, I. Wasif, Physical, chemical and biological treatment of textile wastewater for removal of dyes and heavy metals, *Desalin. Water Treat.* 320 (2024) 100842, <https://doi.org/10.1016/J.DWT.2024.100842>.
- [6] N. Jahan, M. Tahmid, A.Z. Shoronika, A. Fariha, H. Roy, M.N. Pervez, Y. Cai, V. Naddeo, M.S. Islam, A comprehensive review on the sustainable treatment of textile wastewater: zero liquid discharge and resource recovery perspectives, *Sustain* 14 (2022) 1–38, <https://doi.org/10.3390/su142215398>.
- [7] S. Dutta, B. Gupta, S.K. Srivastava, A.K. Gupta, Recent advances on the removal of dyes from wastewater using various adsorbents: a critical review, *Mater. Adv.* 2 (2021) 4497–4531, <https://doi.org/10.1039/D1MA00354B>.
- [8] K.G. Pavithra, S.K. P. V. Jaikumar, S.R. P. Removal of colorants from wastewater: a review on sources and treatment strategies, *J. Ind. Eng. Chem.* 75 (2019) 1–19, <https://doi.org/10.1016/J.JIEC.2019.02.011>.
- [9] M. Guembe-García, G. Utzeri, A.J.M. Valente, S. Ibeas, M. Trigo-López, J.M. García, S. Vallejos, Efficient extraction of textile dyes using reusable acrylic-based smart polymers, *J. Hazard. Mater.* 476 (2024) 135006, <https://doi.org/10.1016/J.JHAZMAT.2024.135006>.
- [10] N.M. Mahmoodi, M.H. Saffar-Dastgerdi, Clean Laccase immobilized nanobiocatalysts (graphene oxide - zeolite nanocomposites): From production to detailed biocatalytic degradation of organic pollutant, *Appl. Catal. B Environ.* 268 (2020) 118443, <https://doi.org/10.1016/J.JAPCATB.2019.118443>.
- [11] K.K.O.S. Silva, C.A. Paskocimas, F.R. Oliveira, J.H.O. Nascimento, A. Zille, Development of porous alumina membranes for treatment of textile effluent, *Desalin. Water Treat.* 57 (2016) 2640–2648, <https://doi.org/10.1080/19443994.2015.1018333>.
- [12] S. Barredo-Damas, M.I. Alcaina-Miranda, A. Bes-Piá, M.I. Iborra-Clar, A. Iborra-Clar, J.A. Mendoza-Roca, Ceramic membrane behavior in textile wastewater ultrafiltration, *Desalination* 250 (2010) 623–628, <https://doi.org/10.1016/j.desal.2009.09.037>.
- [13] A. Arooj, J. Yin, K. Li, M. Li, X. Zhang, Pressure-driven membranes for advanced treatment of textile wastewater: dye/salt fractionation, concentration, and cleaning strategies, *Desalination* 581 (2024) 117596, <https://doi.org/10.1016/j.desal.2024.117596>.
- [14] W. Ye, F. Yu, Z. Yu, N. Kong, X. Lin, R. Liu, J. Du, X. Huang, A. Gu, S. Arcadio, W. Zhang, X. Li, S. Zhao, M. Xie, Q. Liang, J. Lin, High-performance antibacterial tight ultrafiltration membrane constructed by co-deposition of dopamine and tobramycin for sustainable high-salinity textile wastewater management, *Desalination* 579 (2024) 117482, <https://doi.org/10.1016/j.desal.2024.117482>.
- [15] J. Lin, W. Ye, H. Zeng, H. Yang, J. Shen, S. Darvishmanesh, P. Luis, A. Sotto, B. Van der Bruggen, Fractionation of direct dyes and salts in aqueous solution using loose nanofiltration membranes, *J. Memb. Sci.* 477 (2015) 183–193, <https://doi.org/10.1016/J.MEMSCI.2014.12.008>.
- [16] G. Mahmodi, R.R. Bafti, N.I. Boroujeni, S. Pradhan, S. Dangwal, B. Sengupta, V. Zatanpour, M. Sorci, M. Fathizadeh, P. Bikina, G. Belfort, M. Yu, S.J. Kim, Improving cellulose acetate mixed matrix membranes by incorporating hydrophilic MIL-101(Cr)-NH<sub>2</sub> nanoparticles for treating dye/salt solution, *Chem. Eng. J.* 477 (2023) 146736, <https://doi.org/10.1016/j.cej.2023.146736>.
- [17] X. Luo, L. Jiang, R. Zhao, Y. Wang, X. Xiao, S. Ghazouani, L. Yu, Z. Mai, H. Matsuyama, P. Jin, Energy-efficient trehalose-based polyester nanofiltration membranes for zero-discharge textile wastewater treatment, *J. Hazard. Mater.* 465 (2024) 133059, <https://doi.org/10.1016/J.JHAZMAT.2023.133059>.
- [18] M. Liu, Z. Lü, Z. Chen, S. Yu, C. Gao, Comparison of reverse osmosis and nanofiltration membranes in the treatment of biologically treated textile effluent for water reuse, *Desalination* 281 (2011) 372–378, <https://doi.org/10.1016/J.DESAL.2011.08.023>.
- [19] M. Cifuentes-Cabezas, L. García-Suarez, J.L. Soler-Cabezas, B. Cuartas-Urbe, S. Álvarez-Blanco, J.A. Mendoza-Roca, M.C. Vincent-Vela, Feasibility of forward osmosis to recover textile dyes using single salts and multicomponent draw solutions, *Membranes (Basel)* 13 (2023) 1–16, <https://doi.org/10.3390/membranes13120911>.
- [20] J. Korenak, C. Hélix-Nielsen, H. Bukšek, I. Petrinčić, Efficiency and economic feasibility of forward osmosis in textile wastewater treatment, *J. Clean. Prod.* 210 (2019) 1483–1495, <https://doi.org/10.1016/j.jclepro.2018.11.130>.
- [21] A. Altaee, G. Zaragoza, H.R. van Toningen, Comparison between forward osmosis-reverse osmosis and reverse osmosis processes for seawater desalination, *Desalination* 336 (2014) 50–57, <https://doi.org/10.1016/j.desal.2014.01.002>.
- [22] F. Volpin, E. Fons, L. Chekli, J.E. Kim, A. Jang, H.K. Shon, Hybrid forward osmosis-reverse osmosis for wastewater reuse and seawater desalination: Understanding the optimal feed solution to minimise fouling, *Process Saf. Environ. Prot.* 117 (2018) 523–532, <https://doi.org/10.1016/j.psep.2018.05.006>.
- [23] A. Altaee, A. Sharif, G. Zaragoza, A.F. Ismail, Evaluation of FO-RO and PRO-RO designs for power generation and seawater desalination using impaired water feeds, *Desalination* 368 (2015) 27–35, <https://doi.org/10.1016/j.desal.2014.06.022>.
- [24] A. D'Haese, J.C.O. Bravo, D. Harmsen, L. Vanhaecke, A.R.D. Verliefde, D. Jeison, E. R. Cornelissen, Analysing organic micropollutant accumulation in closed loop FO-RO systems: a pilot plant study, *J. Memb. Sci.* 626 (2021) 119182, <https://doi.org/10.1016/j.memsci.2021.119182>.
- [25] A. D'Haese, P. Le-Clech, S. Van Nevel, K. Verbeken, E.R. Cornelissen, S.J. Khan, A. R.D. Verliefde, Trace organic solutes in closed-loop forward osmosis applications: influence of membrane fouling and modeling of solute build-up, *Water Res* 47 (2013) 5232–5244, <https://doi.org/10.1016/j.watres.2013.06.006>.
- [26] P. Liu, X. Yu, Y. Liu, R. Han, H. Li, Simultaneous degradation and rejection of PPCPs from wastewater in a 3DEO-FO-RO process, *Sep. Purif. Technol.* 330 (2024) 125264, <https://doi.org/10.1016/j.seppur.2023.125264>.
- [27] J.M. Bidu, B. Van der Bruggen, M.J. Rwiza, K.N. Njau, Current status of textile wastewater management practices and effluent characteristics in Tanzania, *Water Sci. Technol.* 83 (2021) 2363–2376, <https://doi.org/10.2166/wst.2021.133>.
- [28] A. Khatri, M.H. Peerzada, M. Mohsin, M. White, A review on developments in dyeing cotton fabrics with reactive dyes for reducing effluent pollution, *J. Clean. Prod.* 87 (2015) 50–57, <https://doi.org/10.1016/j.jclepro.2014.09.017>.
- [29] C.M. Sánchez-Arévalo, L. García-Suarez, M.S. Camilleri-Rumbau, J. Vogel, S. Álvarez-Blanco, B. Cuartas-Urbe, M.C. Vincent-Vela, Treatment of industrial textile wastewater by means of forward osmosis aiming to recover dyes and clean water, *Heliyon* 10 (2024), <https://doi.org/10.1016/j.heliyon.2024.e40742>.

- [30] C. Bendicho, I. Lavilla, Water Analysis | Sewage, in: *Encycl. Anal. Sci.*, Elsevier, 2019, pp. 371–381, <https://doi.org/10.1016/B978-0-12-409547-2.11519-7>.
- [31] H. Nabetani, M. Nakajima, A. Watanabe, S. Nakao, S. Kimura, Prediction of the flux for the reverse osmosis of a solution containing sucrose and glucose, *J. Chem. Eng. JAPAN* 25 (1992) 575–580, <https://doi.org/10.1252/jcej.25.575>.
- [32] J. Li, C. Xu, J. Ye, E. Li, S. Xu, M. Huang, Enhanced anti-fouling of forward osmosis membrane by pulsatile flow operation in textile wastewater treatment, *Desalination* 565 (2023) 116878, <https://doi.org/10.1016/j.desal.2023.116878>.
- [33] L. Xia, M.F. Andersen, C. Hélix-Nielsen, J.R. McCutcheon, Novel commercial aquaporin flat-sheet membrane for forward osmosis, *Ind. Eng. Chem. Res.* 56 (2017) 11919–11925, <https://doi.org/10.1021/ACS.IECR.7B02368>.
- [34] B. Yuzer, H. Selcuk, Recovery of biologically treated textile wastewater by ozonation and subsequent bipolar membrane Electrodialysis process, *Membr. Technol.* 2021 11 (2021) 900, <https://doi.org/10.3390/MEMBRANES11110900>. Page11900.
- [35] B.M. Souza-Chaves, M.A. Alhussaini, V. Felix, L.K. Presson, W.Q. Betancourt, K. L. Hickenbottom, A. Achilli, Extending the life of water reuse reverse osmosis membranes using chlorination, *J. Memb. Sci.* 642 (2022) 119897, <https://doi.org/10.1016/J.MEMSCI.2021.119897>.
- [36] R.W. Baker, Reverse osmosis, in: *Membr. Technol. Appl.*, Wiley, 2012, pp. 207–251, <https://doi.org/10.1002/9781118359686.ch5>.
- [37] T.M. Mogashane, J.P. Maree, N. Nyamutswa, J. Vogel, M. Mujuru, Evaluation of forward osmosis for treatment of sodium Sulphate rich brine, in: *SETWM-20, ACBES-20 EEHSS-20 Nov. 16-17, 2020 Johannesburg*, Eminent Association of Pioneers, 2020, <https://doi.org/10.17758/EAREST0.EAP1120243>.
- [38] L. Sbardella, G. Blandin, A. Fàbregas, J. Carlos Real Real, A. Serra Clusellas, F. Ferrari, C. Bosch, X. Martínez-Lladó, Optimization of pilot scale forward osmosis process integrated with electrodialysis to concentrate landfill leachate, *Chem. Eng. J.* 434 (2022), <https://doi.org/10.1016/j.cej.2021.134448>.
- [39] L. Shu, M. Pannirselvam, V. Jegatheesan, Nanofiltration of dye bath towards zero liquid discharge: a technical and economic evaluation, in: M. Pannirselvam, L. Shu, G. Griffin, L. Philip, A. Natarajan, S. Hussain (Eds.), *Water Scarcity Ways to Reduce Impact*, Springer International Publishing, Cham, 2019, pp. 47–61, [https://doi.org/10.1007/978-3-319-75199-3\\_3](https://doi.org/10.1007/978-3-319-75199-3_3).
- [40] L. Wojnárovits, R. Homlok, K. Kovács, A. Tegze, E. Takács, Wastewater Characterization: chemical oxygen demand or total organic carbon content measurement? *Molecules* 29 (2024) <https://doi.org/10.3390/molecules29020405>.
- [41] J.A. Aguilar-Torrejón, P. Balderas-Hernández, G. Roa-Morales, C.E. Barrera-Díaz, I. Rodríguez-Torres, T. Torres-Blancas, Relationship, importance, and development of analytical techniques: cod, BOD, and, TOC in water—an overview through time, *SN Appl. Sci.* 5 (2023) 118, <https://doi.org/10.1007/s42452-023-05318-7>.
- [42] C. Pinto, A. Fernandes, A. Lopes, M.J. Nunes, A. Baía, L. Ciriaco, M.J. Pacheco, Reuse of textile dyeing wastewater treated by electrooxidation, *Water* 2022 14 (2022) 1084, <https://doi.org/10.3390/W14071084>. Page141084.
- [43] S. Ma, X. Wu, L. Fan, Q. Wang, Y. Hu, Z. Xie, Effect of different draw solutions on concentration polarization in a forward osmosis process: theoretical modeling and experimental validation, *Ind. Eng. Chem. Res.* 62 (2023) 3672–3683, <https://doi.org/10.1021/ACS.IECR.2C03723>.
- [44] C. Wei, Y. Lao, R. Ouyang, G. Zhang, G. Huang, F. Deng, Q. Tan, G. Lin, H. Zhou, Evaluation of different reverse osmosis membranes for textile dyeing and finishing wastewater reuse, *Membranes (Basel)* 13 (2023), <https://doi.org/10.3390/membranes13040420>.
- [45] Z. Yin, C. Yang, C. Long, A. Li, Effect of integrated pretreatment technologies on RO membrane fouling for treating textile secondary effluent: Laboratory and pilot-scale experiments, *Chem. Eng. J.* 332 (2018) 109–117, <https://doi.org/10.1016/J.CEJ.2017.09.059>.
- [46] L. Liu, R. Cheng, X. Chen, X. Zheng, L. Shi, D. Cao, Z. Zhang, Applications of membrane technology in treating wastewater from the dyeing industry in China: Current status and prospect, *Desalin. Water Treat.* 77 (2017) 366–376, <https://doi.org/10.5004/dwt.2017.20775>.
- [47] S.J. Im, S. Jeong, S. Jeong, A. Jang, Techno-economic evaluation of an element-scale forward osmosis-reverse osmosis hybrid process for seawater desalination, *Desalination* 476 (2020) 114240, <https://doi.org/10.1016/J.DESAL.2019.114240>.
- [48] L. Wang, B. Xie, H. Hu, G. Bai, Z. He, Y. Huang, L. Liu, C. Meng, Regulating the surface charge and reactivity of cellulose through quaternized modification to achieve salt-free reactive dyeing process, *Int. J. Biol. Macromol.* (2024) 135898, <https://doi.org/10.1016/j.ijbiomac.2024.135898>.
- [49] T.Y. Cath, A.E. Childress, M. Elimelech, Forward osmosis: principles, applications, and recent developments, *J. Memb. Sci.* 281 (2006) 70–87, <https://doi.org/10.1016/j.memsci.2006.05.048>.
- [50] S. Zhao, L. Zou, C.Y. Tang, D. Mulcahy, Recent developments in forward osmosis: opportunities and challenges, *J. Memb. Sci.* 396 (2012) 1–21, <https://doi.org/10.1016/j.memsci.2011.12.023>.
- [51] C. Yang, L. Li, J. Shi, C. Long, A. Li, Advanced treatment of textile dyeing secondary effluent using magnetic anion exchange resin and its effect on organic fouling in subsequent RO membrane, *J. Hazard. Mater.* 284 (2015) 50–57, <https://doi.org/10.1016/J.JHAZMAT.2014.11.011>.
- [52] E. Kadaj, R. Bosleman, Energy recovery devices in membrane desalination processes, *Renew. Energy Powered Desalin. Handb. Appl. Thermodyn.* (2018) 415–444, <https://doi.org/10.1016/B978-0-12-815244-7.00011-8>.
- [53] B.G. Choi, D.I. Kim, S. Hong, Fouling evaluation and mechanisms in a FO-RO hybrid process for direct potable reuse, *J. Memb. Sci.* 520 (2016) 89–98, <https://doi.org/10.1016/J.MEMSCI.2016.07.035>.
- [54] B.G. Choi, M. Zhan, K. Shin, S. Lee, S. Hong, Pilot-scale evaluation of FO-RO osmotic dilution process for treating wastewater from coal-fired power plant integrated with seawater desalination, *J. Memb. Sci.* 540 (2017) 78–87, <https://doi.org/10.1016/J.MEMSCI.2017.06.036>.
- [55] R. Yalamanchili, I. Rodríguez-Roda, A. Galizia, G. Blandin, Can a forward osmosis-reverse osmosis hybrid system achieve 90 % wastewater recovery and desalination energy below 1 kWh/m<sup>3</sup>? A design and simulation study, *Desalination* 585 (2024) 117767, <https://doi.org/10.1016/J.DESAL.2024.117767>.
- [56] E.W. Tow, A.L. Hartman, A. Jaworowski, I. Zucker, S. Kum, M. AzadiAghdam, E. R. Blatchley, A. Achilli, H. Gu, G.M. Urper, D.M. Warsinger, Modeling the energy consumption of potable water reuse schemes, *Water Res.* X 13 (2021) 100126, <https://doi.org/10.1016/J.WROA.2021.100126>.